Prequential Evaluation of Adaptive Prediction Intervals: A Fifteen-Series Replay Study

Felipe Santibáñez-Leal
CAOS open-research programme, Santiago, Chile
fsantibanezleal@ug.uchile.cl

Working revision; not a new published version.

Preprint · v2.6 draft, 5 November 2026 · Licence CC BY 4.0
CAOS open-research programme · CAOS_RES_ChronoScope, time-series forecasting method atlas
Published predecessor DOI: 10.5281/zenodo.23884367 · concept DOI (always latest) 10.5281/zenodo.21508920
Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_RES_ChronoScope

Abstract

Adaptive prediction intervals can have accurate average coverage while retaining substantial local coverage error. We examine this distinction in a one-step prequential study of fifteen selected time series: eleven generated mechanisms and four real excerpts. The point forecaster is a full-history smoothing-plus-drift adaptation, rather than a native stateful forecaster or the standard finite-history Theta method. Two scale-normalized score-buffer controllers use ACI-style and PID-style updates of an effective miscoverage level. At nominal 80% coverage, their mean rolling absolute coverage errors are 0.0656 and 0.0645, compared with 0.0830 for unadjusted intervals. The substantially smaller errors obtained by first averaging signed coverage deviations over time describe a different statistic and do not establish uniformly accurate local coverage. Retrospective paired case resampling gives rolling-error differences of -0.01737 and -0.01851, with exploratory ninety-five-percent intervals [-0.02657,-0.00871] and [-0.02714,-0.01074]. Interval-score differences are small and their intervals include zero. These endpoints describe sensitivity to the weighting of selected cases, not population-wide confidence. Every numerical forecast, target, score and controller update is reconstructed and checked against the archived coverage trajectories and final weighted quantile losses. Sensitivity calculations vary nominal coverage, base forecaster, controller gains and generated-series seeds. These selected finite-stream results characterize the evaluated implementations; they provide neither a new coverage theorem nor a general forecasting advantage.

1 Introduction

A sequential interval forecast must use information available before its target arrives. Its quality has at least two dimensions: coverage and concentration. An interval can obtain high coverage by becoming excessively wide; a long-run coverage average can also conceal sustained local deviations. We ask how two particular adaptive interval controllers behave on a selected collection when these distinctions are made explicit.

Prequential evaluation scores predictions before incorporating their outcomes [24]. Adaptive Conformal Inference (ACI) adapts an effective miscoverage level [25]. Conformal PID Control

combines control-theoretic ideas with prospective conformal score forecasting [26]. The evaluated algorithms build on these precedents. The PID-style controller evaluated here acts on a bounded effective miscoverage level and uses an empirical score buffer. It is not a faithful implementation or comparative evaluation of the published Conformal PID algorithm, and that algorithm's guarantees are not assigned to it.

The contribution is a reproducible, implementation-specific comparison of global and local empirical coverage measures, interval efficiency, and sensitivity on the same streams. The software package `preqts` supplies the replay interface and archived controllers. The experiment compares interval rules; it does not compare native model-state updates or establish a capability gap in other evaluation frameworks.

The distinction matters when a displayed coverage average is used to assess whether uncertainty is reliable. A controller could cover too few outcomes during one regime and too many during another, yet finish near its nominal frequency. The appropriate evidence must reveal the temporal statistic being summarized and the width needed to obtain that coverage. A selected example can explain the mechanism, but it cannot identify the expected performance of a future operational population.

This paper contributes a complete numerical reconstruction of the archived procedure, a derivation of the relation between its coverage summaries, an explicit account of clipping in the feedback equation, and a matched comparison with efficiency and sensitivity. Its new paired analysis is retrospective case-weighting sensitivity, not an independently selected confirmation study. In particular, a smaller coverage error is kept separate from a point-forecast improvement or a guarantee at every time. The title is restricted to adaptive prediction intervals and a fifteen-series replay because those are the objects actually evaluated. The following sections separate prior methods, mathematical properties, experimental reconstruction and the finite comparisons needed to interpret that scope.

2 Related work

2.1 Prospective scoring and conformal validity

Dawid's prequential principle evaluates a forecast through the subsequently observed outcome, with the prediction fixed before that outcome is incorporated [24]. It specifies an information order rather than a particular fitting algorithm. A repeated full-prefix fit and a native incremental state update can both be evaluated prequentially, although their computational costs differ. This study uses the former. Accordingly, an accurate description of its contribution concerns interval evaluation and numerical reconstruction; it cannot establish that existing forecasting tools lack prospective evaluation or that the selected bases are native stateful models.

Conformal prediction provides prediction sets whose validity is tied to an explicit probabilistic or online model. Vovk, Gammerman and Shafer develop the foundational framework [1]; Shafer and Vovk explain both its online interpretation and assumptions [2]. An empirical residual quantile resembles one ingredient of that framework, but the resemblance is insufficient to transfer a theorem. A rolling score buffer, repeatedly refitted point model, clipped nominal level and interpolated quantile must be matched to the assumptions and exact construction of a proposed result. The present replay documents those choices rather than assigning exchangeable-data validity to a serially dependent experiment.

Lei and colleagues analyze full and split conformal regression, including efficiency and locally varying interval length [3]. Their distinction between fitting and calibration information helps explain why in-sample Gaussian residual dispersion is not an independent calibration sample. Romano, Patterson and Candès combine quantile regression with conformal calibration [4]. Their

construction adapts interval shape through estimated conditional quantiles; the replay instead uses symmetric intervals around a fixed point base and rescales them using past normalized errors. These are different procedures. No conformalized quantile-regression comparator is run here, so its published results are context rather than a numerical baseline in the tables.

Marginal coverage also differs from conditional coverage. Barber and colleagues formalize limitations of unrestricted distribution-free conditional inference [5]. That result cautions against interpreting a marginal empirical frequency as validity at every time, state or covariate value. It does not excuse avoidable empirical undercoverage or imply that useful conditional improvements are impossible with additional assumptions. The present rolling statistic describes local finite-stream behavior. It is neither a test of every possible conditioning event nor an estimator with an automatically valid confidence band under arbitrary dependence.

2.2 Dependence and changing distributions

Several established methods explicitly address departures from exchangeability. Chernozhukov, Wüthrich and Zhu use structured randomization to accommodate dependent observations, with approximate validity under stated conditions when exchangeability fails [6]. Tibshirani and colleagues use weighted conformal prediction for covariate shift, with a known or estimated likelihood ratio [7]. Barber and colleagues further study robustness beyond exchangeability and nonsymmetric fitting procedures [8]. These approaches introduce assumptions or constructions beyond merely keeping recent scores. The present study estimates no density ratio and runs no block-randomization procedure. Its finite-memory controller should therefore not be presented as an interchangeable implementation of those methods.

Xu and Xie's EnbPI combines bootstrap ensemble predictions and sequential residual information for dynamic time series, with approximate coverage results under the paper's conditions [9]. Their later sequential predictive conformal inference predicts a future residual quantile by exploiting temporal score dependence [10]. The replay shares an interest in sequential residuals but does not fit an ensemble or a conditional residual-quantile model. Its buffer takes an unconditional interpolated quantile at an adaptively chosen level. EnbPI and SPCI are consequently relevant future comparators, and their omission prevents a claim of superiority over established time-series conformal procedures.

Gibbs and Candès introduce ACI to adapt the effective miscoverage parameter under distribution shift [25]. The replay uses that feedback direction: a miss lowers the effective miscoverage, which usually raises a buffered residual quantile and widens the next interval. However, the replay clips this parameter and uses a particular finite buffer and numerical quantile convention. Those choices define an ACI-style variant rather than proving identity with every setting analyzed in the original paper. The derivation below exposes the additional clipping term in its cumulative-error equation.

Zaffran and colleagues analyze learning-rate effects and propose AgACI through online aggregation [11]. Gibbs and Candès subsequently study online conformal prediction with arbitrary distribution shifts and adaptive step-size selection [12]. Both motivate treating gain choice as a substantive design component. The replay retains inherited fixed gains, and its multiplier grid is a sensitivity analysis. It neither implements their adaptive aggregation nor selects a gain on independent validation data. Consequently, reporting the best cell of this grid as a validated improvement would answer a different and unsupported question.

2.3 Local adaptation, control and forecast evaluation

Bhatnagar and colleagues develop strongly adaptive online conformal methods whose regret analysis concerns changing local intervals rather than only one whole-horizon average [13]. This distinction is directly relevant to the difference between averaging signed coverage deviations and averaging their absolute values. Nevertheless, the study's rolling absolute error is a descriptive statistic with a twenty-five-arrival window, not the strongly adaptive regret objective or a reproduction of that paper's algorithms. A lower displayed rolling error cannot establish the corresponding theoretical guarantee.

Angelopoulos, Barber and Bates analyze online conformal prediction with decaying step sizes, distinguishing retrospective coverage from convergence toward a population quantile in stable settings [14]. The replay has fixed gains and does not estimate the step-size schedule of that method. Angelopoulos, Candès and Tibshirani's Conformal PID Control combines online calibration with prospective score forecasting [26]. The historical code considered here instead applies proportional, accumulated and difference terms to a clipped effective miscoverage level. Its PID-style name records that implementation pattern, not a faithful evaluation of the published algorithm. Neither publication's guarantees are imported by naming the controller.

Forecast evaluation must also account for concentration. Gneiting, Balabdaoui and Raftery distinguish calibration from sharpness [15], and Gneiting and Raftery develop proper scoring-rule principles [28]. The interval score used here penalizes both width and excluded outcomes. It can reveal that a coverage improvement is accompanied by little improvement in the interval's overall score. This is particularly relevant when every controller retains the same point forecast. The study consequently reports scaled interval score and width rather than presenting coverage as a complete measure of probabilistic forecasting quality.

Finally, forecasting evaluation depends on origin choice, information availability and aggregation. Tashman's analysis of out-of-sample evaluation motivates retaining temporal order and explicit evaluation schedules [16]. Hyndman and Koehler's discussion of error measures motivates scrutiny of scaling and zero denominators [17]. The replay uses a warmup-only seasonal scale for cross-case interval comparisons and retains the historical per-arrival quantile-loss ratio only for archive reconciliation. These decisions prevent a later outcome from setting the scale and expose the unstable behavior of a denominator near zero. They do not establish that the selected fifteen streams represent an operational population or that their average score identifies a generally optimal interval controller.

3 Problem formulation, data and sequential information

The collection contains fifteen deliberately selected cases: eleven generated mechanisms and four real excerpts. Generated mechanisms include daily seasonality, trend with seasonality, intermittent demand, a near-random walk, white noise, level changes, multiple seasonalities, heteroskedasticity, long memory, nonlinear Mackey–Glass dynamics and a promotion covariate example. Real excerpts comprise UCI electricity, Beijing PM2.5, M4 hourly and M4 daily. They do not constitute random draws from a forecasting population. White noise and the near-random walk are negative controls, not proofs that every form of leakage is absent.

Table 1 gives every stream length, warmup, seasonal setting and scored-arrival count. The authentic inputs are admitted excerpts rather than complete benchmark inventories: UCI ElectricityLoadDiagrams20112014 [21], UCI Beijing PM2.5 [22], and the Monash archive's M4 daily and hourly collections [23]. The UCI records and the admitted provenance specify CC BY 4.0. Generator definitions and exact numerical source bindings remain part of the replay evidence.

Table 1: Complete replay inventory with original sequence length, initial prefix, configured seasonal lag and scored arrivals; generated cases and authentic excerpts remain distinct provenance classes.

Case	T	w	m	Arrivals	Provenance
BRKV_level_shift	300	24	12	276	Generated
CHAO_mackey	500	20	1	480	Generated
CTRL_white_noise	160	20	1	140	Generated
EXOG_promo	240	20	7	220	Generated
HETV_garch	500	20	1	480	Generated
INTM_demand	200	20	1	180	Generated
LMEM_fractional	500	20	1	480	Generated
MSEA_daily_weekly	672	48	24	624	Generated
REAL_electricity	480	48	24	432	UCI
REAL_m4_daily	400	20	7	380	Monash
REAL_m4_hourly	480	48	24	432	Monash
REAL_pm25	480	48	24	432	UCI
RWLK_noise	200	20	1	180	Generated
SEAS_hourly	480	48	24	432	Generated
TRND_seasonal	180	24	12	156	Generated

The historical electricity loader sums four quarter-hour power readings for each delivered hour. That sum is four times their arithmetic mean, not a conversion to hourly energy. The replay preserves those admitted numbers and treats their source unit explicitly; it does not infer an energy balance or operational capacity from them. The PM2.5 loader forward-fills missing values in its selected window, with an initial fallback of zero if the first retained value is missing. Those are preprocessing policies, not newly measured observations. The current dataset record attributes Beijing PM2.5 to Song Chen; the historical manifest's Liang citation refers to associated work and is not used as a replacement for the dataset citation here.

The historical M4 excerpt loader chooses the first sufficiently long cached series with nontrivial variance and a lag correlation of at least .2 over the candidate excerpt. This is deliberate retrospective case selection using the excerpt, including outcomes later scored by the stream. It does not give those future values to an earlier point fit, but it does limit any claim that the cases were independently or prospectively selected. The paper preserves that distinction rather than treating these curated examples as a random benchmark sample.

Let $y_0, \dots, y_{T-1}$ be the complete stream and m its configured seasonal lag. The initial prefix has length

$$w = \min\{\max(2m, 20), \max(1, \lfloor T/3 \rfloor)\}. \tag{1}$$

At $t = w, \dots, T-1$, every method predicts y_t using $y_{0:t-1}$. The target is then revealed, the forecast is scored, and the prefix and controller are updated. Horizon and advance are both one. This streaming experiment includes the full case sequence; its targets must not be pooled with the separate final-block experiment in the surrounding atlas.

Figure 1 distinguishes the prefix-only fit, interval issuance, target revelation and the two subsequent updates; its arrows encode information availability rather than a claim about native state-update performance.

The numerical reconstruction retains the full-precision source values, generator definitions and seeds, case identities, seasonal lags and initial-prefix lengths. SHA-256 values bind each source array, admitted real-data CSV, generator implementation, case manifest, display trace and archived streaming artifact. Every displayed sample and reserved final-block target is matched to the original indexed source within its four-decimal export tolerance. These bindings identify the actual experimental inputs, even where a historical upstream download revision or timing

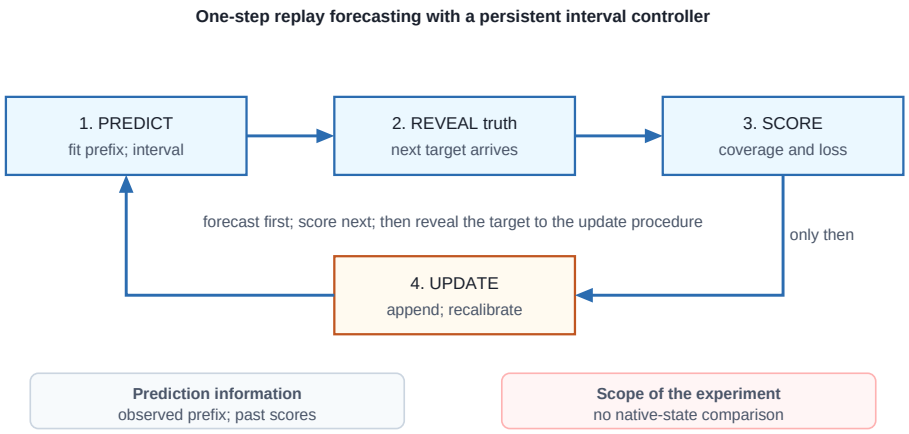


Figure 1: Information order in the evaluated replay implementation. The point method is recomputed on the observed prefix; an interval controller carries its own scores and level between arrivals. The next target enters either update only after scoring. This diagram makes no comparison with other forecasting frameworks or native state-update costs.

environment is unavailable. Covariates are not supplied to the four interval methods, including for the promotion case; no covariate-availability result is inferred here.

3.1 Notation, units and source identity

Table 2 fixes the quantities used by the replay. The unit U denotes the source measurement unit; it is not assigned an SI interpretation when the admitted excerpt lacks one. Seasonal lags and warmup lengths count observations on that source's supplied grid. Equal counts across two cases do not imply equal physical durations. The study does not resample every real excerpt to a common temporal unit or claim that its lag choices were optimized for the controller comparison.

Source identity is distinct from statistical representativeness. A digest of the complete float64 source array identifies the values reconstructed for this experiment. The manifest and raw admitted CSV bindings add provenance, but a hash does not recover an unavailable upstream revision or prove the excerpt is representative. Likewise, a generated seed identifies one realization within a known mechanism, not a newly collected real-world dataset. The two additional seeds retain each mechanism's definition and seasonal setting. They are discussed separately from the four authentic excerpts throughout the analysis.

The distinction between observation units and dimensionless controller state matters when reproducing a numerical floor. A base half-width is in source units, while a normalized error, empirical score multiplier and effective miscoverage are dimensionless. Scaling all observations should therefore preserve the controller sequence only when the source-unit floor is inactive. The derivation states that condition explicitly rather than treating the historical floor as a universal physical constant.

4 Replay method and controller derivation

4.1 Point forecasts and base intervals

We call the archived adaptation smoothing plus drift (S+D). The historical artifact key `Theta` identifies this implementation, but it lacks the finite-history correction of the standard nonseasonal

Table 2: Notation for the one-step replay, normalized residual buffer, feedback state and finite-stream evaluation, with source-unit quantities distinguished from probabilities and observation counts.

Symbol	Meaning	Unit
y_t, p_t, l_t, u_t	Target, point forecast and interval endpoints.	U
T, n, t, w, m	Stream length, prefix length, index, warmup and seasonal lag.	observations
a, b, ℓ_i	Smoothing coefficient, fitted drift and level.	1; U/observation; U
c, α^*, α_t	Nominal coverage, target and effective miscoverage.	dimensionless
$\hat{\sigma}_t, r_t$	Fitted dispersion and base half-width.	U
$z_t, q_t, \mathcal{B}_t$	Normalized error, buffered quantile and prior score buffer.	dimensionless
e_t, B_t, g_t	Miss, hit and signed feedback error, where stated.	dimensionless
$I_t, d_t, \gamma, k_p, k_i, k_d$	Accumulated error, difference error and gains per arrival.	dimensionless
N, K_i, W	Case count, scored-arrival count and rolling window.	counts
C_{it}, E_{avg}, E_{roll}, E_{final}	Rolling coverage and the three finite-stream errors.	dimensionless
δ_t, ω_s	Projection residual and rolling-average boundary weight.	dimensionless
IS_t, s_i	Interval score and initial-prefix seasonal scale.	U
$L_{i,A}, d_i, \widehat{\Delta}$	Case loss, paired difference and mean difference.	dimensionless

Theta representation [27]. For the observed prefix $x_0, \ldots, x_{n-1}$, an OLS line against integer time gives slope b. A smoothing coefficient a is selected from $\{.05, .10, \ldots, .95\}$ by minimum in-sample one-step squared error. With $\ell_0 = x_0$,

$$\ell_i = ax_i + (1-a)\ell_{i-1}, \qquad p_t = \ell_{n-1} + \tfrac{1}{2}b. \tag{2}$$

The fitted value for x_i is ℓ_{i-1}; $\hat{\sigma}_t$ is the population standard deviation of these residuals for $i = 1, \ldots, n-1$. At nominal coverage c, the base half-width is $\Phi^{-1}((1+c)/2)\hat{\sigma}_t$, using the package's Acklam normal-quantile approximation. This Gaussian approximation has no distribution-free coverage guarantee.

A second base is seasonal naive, $p_t = y_{t-m}$, with dispersion equal to the population standard deviation of observed lag-m differences. The historical comparison contains seasonal naive and raw S+D, plus two calibrated versions of S+D. The sensitivity comparison applies the same controllers to both bases. Both are full-prefix replay calculations; the point forecasts do not become streaming-native merely because the interval controller has persistent state.

Simple exponential smoothing supplies a level recurrence, while an added fitted drift changes its extrapolation [20]. Selecting a coefficient on fitted residual error is a deterministic in-prefix optimization over nineteen candidates. It is not selection on an independent validation set, and its residual dispersion can be optimistic. The historical coefficient grid and drift term are kept unchanged so that a numerical discrepancy cannot be disguised as a methodological improvement. The corrected finite-history Theta used elsewhere in the application has a different expression and is not substituted here under the same artifact key.

The base-interval construction assumes that fitted residual dispersion gives a useful future scale. Serial dependence, an abrupt level change and heteroskedasticity can violate that assumption. A seasonal-naive dispersion estimates differences at the configured lag rather than the same smoothing residual distribution. These choices explain why the sensitivity design includes both bases. They also bound the scientific conclusion: a controller's performance around one inaccurate scale approximation does not establish its behavior around every possible point and dispersion model.

4.2 Adaptive interval rules

For a base point forecast p_t and half-width $r_t = \max(\Phi^{-1}((1+c)/2)\hat{\sigma}_t, 10^{-8})$, a FIFO buffer retains at most 200 past scores

$$z_j = |y_j - p_j|/r_j, \qquad j < t. \tag{3}$$

Before forecasting y_t, q_t is the buffered sample quantile at level $1-\alpha_t$, using NumPy's default linear interpolation. An empty buffer gives $q_t = 1$. Endpoints are $p_t \pm q_t r_t$. Only after scoring is z_t appended. This is a normalized-residual buffer rule, not a finite-sample split-conformal construction.

Let $e_t = \mathbf{1}\{z_t > q_t\}$, $\alpha^* = 1 - c$, and initialize $\alpha_w = \alpha^*$. The ACI-style update is

$$\alpha_{t+1} = \text{clip}(\alpha_t + \gamma(\alpha^* - e_t), 10^{-4}, 1 - 10^{-4}), \tag{4}$$

with $\gamma = .02$. Level clipping and the interpolated finite score buffer are part of this evaluated variant; the ACI paper's theoretical guarantee is not claimed for it without further argument.

For the PID-style rule, set $g_t = \alpha^* - e_t$, $I_t = I_{t-1} + g_t$ with initial integral zero, and $d_t = g_t - g_{t-1}$, using zero derivative on the first update. Then

$$\alpha_{t+1} = \text{clip}(\alpha^* + k_p g_t + k_i I_t + k_d d_t, 10^{-4}, 1 - 10^{-4}), \tag{5}$$

where $(k_p, k_i, k_d) = (.05, .02, .02)$. The integral is not separately clipped. Unlike the ACI recursion, this rule recomputes the level around α^* from all three terms. The median is unchanged by either controller.

The gains are inherited fixed software settings. No historical parameter-selection rationale, optimality study or independent tuning set is available. Sensitivity uses all multipliers $g \in \{.5, 1, 2\}$, multiplying γ or all three PID gains together. It does not choose and re-evaluate the best multiplier. Nominal coverages are .7, .8, .9. The full design gives $15 \times 2 \times 3 \times (1+3+3) = 630$ case-setting rows, including raw references.

4.3 Algorithm, information ownership and computational cost

Algorithm 1 specifies the reconstructed order. The point fit and its residual dispersion are recomputed from the entire observed prefix. The controller alone keeps a bounded score buffer and its feedback state. Two implementations that generate the same point sequence can therefore have very different fitting costs. With nineteen smoothing candidates, a prefix of length n costs $O(19n)$ arithmetic operations for the candidate recurrences plus a linear regression for drift. Repeating that calculation over an expanding stream costs quadratically in its length. No native state-update advantage is inferred from this replay implementation.

The controller's numerical work is bounded by a two-hundred-entry buffer. Computing an interpolated empirical quantile requires processing that buffer; a straightforward sorted implementation costs $O(B \log B)$, while a selection-based routine can differ. The constant bound $B = 200$ prevents this part from growing with stream length. These operation counts explain implementation structure and do not replace measured cold-start, fitting or inference latency. Original timing-environment details are missing, and numerical reconstruction time on a new machine is a separate measurement.

The first forecast uses an empty buffer and multiplier one. This is an initialization policy, not a calibrated prediction set obtained from two hundred previous errors. Early rolling coverage also averages fewer than twenty-five arrivals. Both initialization effects remain in the reported stream rather than being removed after inspecting their performance. A different burn-in rule could

Algorithm 1 Reconstruct one adaptive interval stream while keeping its target outside the prediction and score-buffer quantile until the observation step.

Require: Source sequence, seasonal lag, warmup, nominal coverage, base and fixed gains

Initialize empty score buffer, effective miscoverage, integral and previous error
for each cutoff after warmup **do**
    Fit the declared point base and dispersion on the observed prefix only
    Form the base half-width and the past-buffer quantile at the current level
    Fix and record the point forecast and interval endpoints
    Reveal the next target; compute hit, normalized score and proper interval score
    Append the score, removing the oldest entry when capacity is exceeded
    Apply the declared ACI-style or PID-style update and record clipping and state
    Extend the observed prefix with this target
end for
Retain every row, source identity and setting before computing case summaries

change the numerical comparison and would have to be stated as another experiment. Likewise, retaining the most recent two hundred scores favors adaptation to recent history while discarding older outcomes; it is not an independently selected optimal memory length.

4.4 Causality and invariance of the point forecast

Let $\mathcal{F}_{t-1}$ contain the observed prefix and the fixed configuration. Assume the initial controller state is fixed, the base-fitting routine is deterministic for its declared inputs, and the buffer contains only scores from cutoffs before t. Then the point, dispersion, current level and buffered quantile are functions of $\mathcal{F}_{t-1}$. Therefore

$$(p_t, l_t, g_t, \alpha_t, \mathcal{B}_t) \text{ are } \mathcal{F}_{t-1}\text{-measurable.} \tag{6}$$

After the observation arrives, its score and miss indicator become functions of $\mathcal{F}_t$. Appending that score and applying either stated controller makes the next state a function of $\mathcal{F}_t$. Induction from the fixed initial state proves the information-order property at every cutoff. The argument establishes absence of look-ahead in this specified procedure; it does not establish that a different data-loading or preprocessing pipeline respects the same information boundary.

The normalization divides the absolute point error by the base half-width before buffering it. The point fit does not consume the buffer or controller level. For a fixed base and prefix, it follows directly that

$$p_t^{\text{raw}} = p_t^{\text{ACI}} = p_t^{\text{PID}}. \tag{7}$$

Thus every point-error statistic is identical across these interval rules, apart from an implementation defect. The additional analysis checks this identity over every retained row before resampling cases. A reduction in rolling coverage error is therefore an improvement in the interval procedure's measured behavior, not evidence of a more accurate point model. Seasonal naive is a different point base and is interpreted separately.

Positive affine changes of the source also clarify what is normalized. Away from the absolute half-width floor, replacing y by $a + by$, $b > 0$, transforms the point and base width to $a + bp$ and br. The score remains $|b(y - p)|/(br)$, so the controller's score sequence, hits and levels remain unchanged. Interval score scales by b, and dividing it by a positive warmup difference scale cancels that factor. The floor of 10^{-8} has source units and breaks exact scale equivariance when active. It is retained as a historical numerical choice, not described as dimensionless or silently removed from the reconstruction.

4.5 What the two coverage averages measure

Write C_{it} for a case's trailing empirical coverage and use the same fixed nominal coverage c at all its cutoffs. The absolute value is convex. Applying the triangle inequality within each case and then averaging gives the deterministic relation

$$E_{\text{avg}} = \frac{1}{N}\sum_i \left|\frac{1}{K_i}\sum_t (C_{it} - c)\right| \le \frac{1}{N}\sum_i \frac{1}{K_i}\sum_t |C_{it} - c| = E_{\text{roll}}. \tag{8}$$

This inequality needs no independence assumption and holds for every finite realized set of streams. It explains why the smaller statistic cannot substitute for local absolute error. Opposite-signed deviations can cancel before the absolute value is taken. The inequality does not guarantee that either statistic is small, and a small finite rolling error does not imply conditional validity under an unknown future distribution.

Even the average of rolling coverages is not exactly the ordinary final empirical coverage. Index a case's scored arrivals by $s = 1, \ldots, K$ and let $W = 25$. Then

$$\frac{1}{K}\sum_{t=1}^{K} C_t = \sum_{s=1}^{K} \omega_s B_s, \tag{9}$$

$$\omega_s = \frac{1}{K}\sum_{t=s}^{\min(K, s+W-1)} \frac{1}{\min(W, t)}, \qquad \sum_s \omega_s = 1. \tag{10}$$

To verify the formula, expand each rolling mean and interchange the two finite sums. An arrival appears only in subsequent windows that still include it. Initial windows have shorter denominators, while final arrivals appear in fewer windows before the stream ends. The resulting boundary weights differ from $1/K$. Consequently, E_{avg} and E_{final} can differ even before considering the outer absolute value. The reconstruction retains the archive's five-decimal rolling values; the algebra describes the corresponding unrounded quantity, with export rounding supplying a small separate numerical discrepancy.

These results identify precisely the source of the earlier headline ambiguity. They do not identify a universally preferable coverage measure. Final coverage answers a whole-stream frequency question; rolling absolute error describes a chosen local window; a strongly adaptive or conditional criterion addresses another question under its own definitions. This paper uses the first two descriptively and does not claim the latter from a favorable average.

4.6 Clipping and the limits of a cumulative feedback argument

The unprojected ACI-style recursion has an elementary telescoping identity. Let $g_t = \alpha^* - e_t = B_t - c$. Define the projection residual

$$\delta_t = \alpha_{t+1} - \alpha_t - \gamma g_t. \tag{11}$$

Summing over K scored arrivals gives

$$\frac{1}{K}\sum_t (B_t - c) = \frac{\alpha_{\text{end}} - \alpha_{\text{start}}}{\gamma K} - \frac{1}{\gamma K}\sum_t \delta_t. \tag{12}$$

This is an exact algebraic identity for the stated update. If no clipping occurs, every projection residual is zero. Bounded levels then bound the first term by a constant divided by γK. When clipping does occur, the second term is present. A bound on the level's range alone cannot remove it. This distinction prevents the paper from importing a long-run result proved for a

different update or for a construction that permits empty or unbounded prediction sets outside the usual probability range.

A simple limiting example makes the issue concrete. Suppose a base predicts a constant stream exactly, its positive numerical half-width is retained, and all normalized errors are zero. After the first observation the empirical quantile is zero, the interval equals the target, and every arrival is covered. The effective level increases until it reaches the upper clipping boundary. At that boundary the projection residual cancels the proposed positive increment. The coverage frequency remains one rather than converging to $c < 1$. This is a counterexample to an unconditional arbitrary-sequence coverage claim for the clipped finite-buffer construction. It is not a claim that this degeneracy occurred in the fifteen admitted cases, and its zero warmup difference scale would make the cross-case scaled score unavailable.

The PID-style recursion has an accumulated error but a different state equation: it recomputes the level around α^* instead of adding a correction to the previous level. Its unconstrained integral can grow while the effective level is clipped. The ACI telescoping argument therefore cannot be transferred to it by replacing the gain. Proving stability, efficiency or conditional coverage would require assumptions and analysis of that actual coupled system. The empirical sensitivity grid measures behavior at three gain multipliers; it does not supply such a proof. The defensible theoretical claims here are the information-order property, point identity, deterministic coverage inequality and explicit feedback identity, each with its stated conditions.

5 Experimental design, metrics and numerical reconstruction

For case i, write $B_{it} = \mathbf{1}\{l_{it} \le y_{it} \le u_{it}\}$ and let C_{it} be its trailing coverage mean over at most 25 cutoffs, including t. With $N = 15$ cases and K_i cutoffs, three different errors are

$$E_{\text{avg}} = N^{-1} \sum_i \left| K_i^{-1} \sum_t C_{it} - c \right|, \tag{13}$$

$$E_{\text{roll}} = N^{-1} \sum_i K_i^{-1} \sum_t |C_{it} - c|, \tag{14}$$

$$E_{\text{final}} = N^{-1} \sum_i \left| K_i^{-1} \sum_t B_{it} - c \right|. \tag{15}$$

E_{avg} permits temporal cancellation within each case; E_{roll} does not. Neither is a conditional-coverage guarantee, and overlapping rolling windows are not independent observations. Archived C_{it} values are rounded to five decimals; the reproduction uses that same rounding for these comparisons. Equal case weights prevent longer streams from dominating the summary.

Width is $u_t - l_t$. The central interval score is

$$\text{IS}_t = (u_t - l_t) + \frac{2}{1-c}\{(l_t - y_t)_+ + (y_t - u_t)_+\}, \tag{16}$$

where only the violated side contributes [28]. For cross-case summaries, width and interval score are divided by the mean absolute lag-m difference in that case's initial prefix. This is the same fixed information set used for scaling one-step absolute error; final targets do not determine the scale.

At levels $q \in \{.1, .5, .9\}$, the archived WQL is $2 \sum_q \rho_q(y_t - Q_{tq})/|y_t|$, where $\rho_q(v) = v(q - \mathbf{1}\{v < 0\})$. It is unavailable when $y_t = 0$. This per-cutoff ratio is not a pooled all-time denominator or a CRPS estimate. Its archive comparison averages available cutoff values. We do not optimize or headline this ratio, whose denominator can be unstable near zero.

The reconstruction exports 21,296 method-cutoff rows, each containing the target, three forecast columns, normalized score, multiplier, effective level before and after update, integral, derivative and buffer size. Previous score rows determine the entire FIFO contents. All fifteen archived coverage trajectories and final WQL summaries agree at their saved precision. Unit checks compare controller forecasts and levels directly with `preqts==0.3.0` and verify that changing a future target cannot alter an earlier forecast. A separate clean-process recomputation checks the complete numerical payload. The reconstruction is not an independent statistical replication, and it does not recover historical hardware or per-call latency.

5.1 Design, comparisons and inferential units

The historical experiment was not preregistered. Its selected cases, point base and controller settings were inherited from the released implementation. The reconstructed study therefore asks retrospective questions: whether the recorded controllers reduce each stated coverage error on those same streams; how width-sensitive scores change; whether conclusions persist across the fixed coverage and gain grid; and how much the generated mechanisms change at two additional seeds. These questions organize the evidence without pretending that the original settings were chosen through a prospectively documented validation study.

All compared interval rules observe the same target at the same cutoff. They use the same warmup, seasonal lag and nominal coverage within a setting. The raw S+D reference, its ACI-style controller and its PID-style controller share an identical point path. This pairing removes target and point-prediction differences from their interval comparison. Seasonal naive provides another base, not a fourth interval treatment on the S+D point path. The sensitivity grid repeats the interval treatments around each base so that the point-base choice remains visible rather than confounded with the gain multiplier.

The statistical unit for a cross-case summary is one complete stream. There are 5,324 scored cutoffs across the original fifteen cases and four methods per cutoff, producing 21,296 method-cutoff records. These records do not constitute 21,296 independent replications. Arrivals share histories, controller state and overlapping rolling windows; paired methods additionally share each target. Equal-case weighting answers a question about the selected inventory. Equal-arrival weighting would answer a different question and allow longer streams to dominate. Neither weighting scheme establishes representative sampling of an operational population.

The new retrospective extension fixes its resampling settings in a committed analysis plan before computing that extension, while explicitly recording that the original outcomes were already known. It uses ten thousand PCG64 draws with seed 20261105. Each draw samples fifteen complete case identities with replacement and applies the same identities to all methods and both metrics. Point forecasts, targets and within-case trajectories are never resampled separately. The paired difference for a controller A and raw reference R is

$$d_i = L_{i,A} - L_{i,R}, \qquad \widehat{\Delta} = \frac{1}{15}\sum_{i} d_i, \qquad \widehat{\Delta}^{*} = \frac{1}{15}\sum_{j=1}^{15} d_{I_j}. \tag{17}$$

The reported interval uses the 2.5 and 97.5 percentiles of the resampled means. Bootstrap resampling originates in empirical-distribution inference [18]; dependence within a time series requires additional treatment when individual observations are resampled [19]. Here the full-case draw preserves within-stream dependence but does not remove dependence between source cases or create a random sampling design. The endpoints are consequently presented as exploratory case-reweighting sensitivity, not a population coverage guarantee. There is no multiplicity adjustment, no gain selection from these endpoints and no new independent confirmation.

5.2 Reconstructing the numerical state at an arrival

The evidence distinguishes the information used to issue an interval from state written after scoring it. Each row contains the original cutoff, actual target, lower endpoint, point, upper endpoint, normalized score, buffered multiplier, effective levels before and after the update, accumulated error after the update, difference term, buffer size after insertion, hit indicator and historical quantile-loss value. These fields permit a direct audit of a controller transition. They are not interchangeable snapshots: using the after-update level to recreate the same row's forecast would allow its outcome to change its own interval.

For a controlled stream, the score buffer before its first scored arrival is empty. At the jth scored arrival, indexed from zero, its size before forecasting is $\min(200, j)$ and its size after insertion is $\min(200, j + 1)$. The prior buffer can be reconstructed by taking at most the most recent two hundred normalized scores from earlier rows, in order. The current row's score is excluded. Its saved multiplier should then agree with the default linear-interpolation quantile at one minus the saved before-update level, except for the declared empty-buffer multiplier of one. Raw references do not append controller scores, so their saved controller-buffer count remains zero.

The before-update effective level also has a cross-row constraint. The first value is the target miscoverage; every later value equals the preceding after-update level. The PID-style accumulated error is reconstructed from the target-minus-miss sequence, with a zero initial integral and zero initial difference. These checks distinguish a correct numerical recurrence from an artifact that contains plausible-looking but inconsistent state columns. They also make clipping observable without treating a clipped value as evidence of stability.

Source reconstruction is a separate prerequisite. The stream uses all original ordered observations, whereas the browser display trace can retain only indexed samples from a longer history. The audit matches those samples at their original positions and verifies reserved targets at the export tolerance; it never treats separated display samples as consecutive arrivals. Full-precision source arrays and exact byte bindings identify the calculation. Their preservation addresses the earlier missing per-update evidence, while the unavailable original timing environment and some upstream acquisition revisions remain disclosed limitations. Reconstructing these numerical states is a reproducibility improvement, not a new statistical replication.

6 Results

Table 3 compares all three coverage summaries; Table 4 retains width and the proper interval score on the same streams. At nominal 80%, raw S+D has $E_{\text{roll}} = .0830$, compared with .0656

Table 3: Coverage errors on the same fifteen cases at nominal 80%. The three columns are distinct statistics defined in the text; none is a confidence interval.

Method	E_{avg}	E_{roll}	E_{final}
Seasonal naive	0.0599	0.1309	0.0620
S+D	0.0416	0.0830	0.0444
S+D + ACI-style	0.0057	0.0656	0.0075
S+D + PID-style	0.0060	0.0645	0.0074

for the ACI-style rule and .0645 for the PID-style rule. This is a descriptive reduction of roughly 21% and 22% on these cases. By contrast, E_{avg} falls from .0416 to .0057 and .0060. The sevenfold ratio for the latter statistic does not describe local rolling error. Reporting these quantities under a single heading of calibration error would conceal their different interpretations.

Figure 2 displays the two differently ordered averaging operations, while Figure 3 shows the

Table 4: Interval width and interval score divided by each case's warmup seasonal-difference scale, then averaged equally over cases. Lower interval score is better; smaller width alone is not.

Method	Scaled width	Scaled score
Seasonal naive	4.2454	6.7012
S+D	2.8403	4.0886
S+D + ACI-style	2.7574	4.0808
S+D + PID-style	2.7855	4.0834

range of final empirical frequencies across cases.

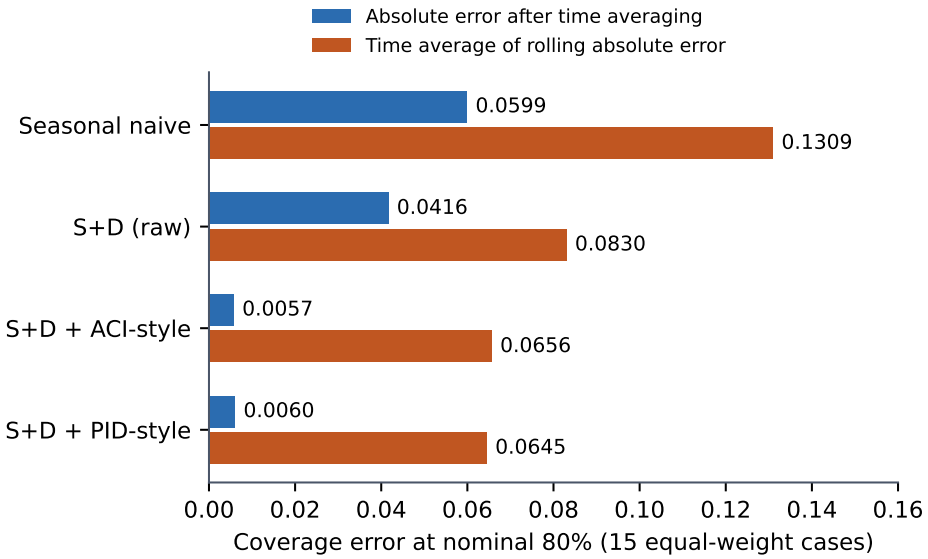


Figure 2: Two coverage-error summaries on the same fifteen streams. Time-averaging before taking absolute deviations permits cancellation; averaging rolling absolute deviations measures a different property. Both remain descriptive finite-stream measurements.

The median forecast is identical across the three S+D interval rules by construction. Thus changes in interval quality do not demonstrate improved point forecasting. Scaled interval widths and scores quantify the accompanying concentration tradeoff, rather than treating coverage alone as sufficient.

Table 5 adds paired case differences for rolling coverage error and warmup-scaled interval score. Figure 4 displays the same differences with the exploratory resampling endpoints. Negative values favor the controller under the named metric. The figures use one source-bound numerical analysis, rather than introducing an additional experiment or changing the original forecasts.

Table 5: Paired differences from raw smoothing plus drift and exploratory ninety-five-percent case-reweighting intervals; entire selected cases, not individual arrivals, are resampled jointly.

Controller	Quantity	Difference	Reweighting interval	Improved cases
ACI-style	Rolling error	-0.01737	[-0.02657, -0.00871]	13/15
ACI-style	Scaled score	-0.00778	[-0.08533, 0.05794]	5/15
PID-style	Rolling error	-0.01851	[-0.02714, -0.01074]	14/15
PID-style	Scaled score	-0.00517	[-0.10092, 0.07694]	6/15

The ACI-style rolling-error difference is -0.01737, with case-reweighting interval [-0.02657,-0.00871].

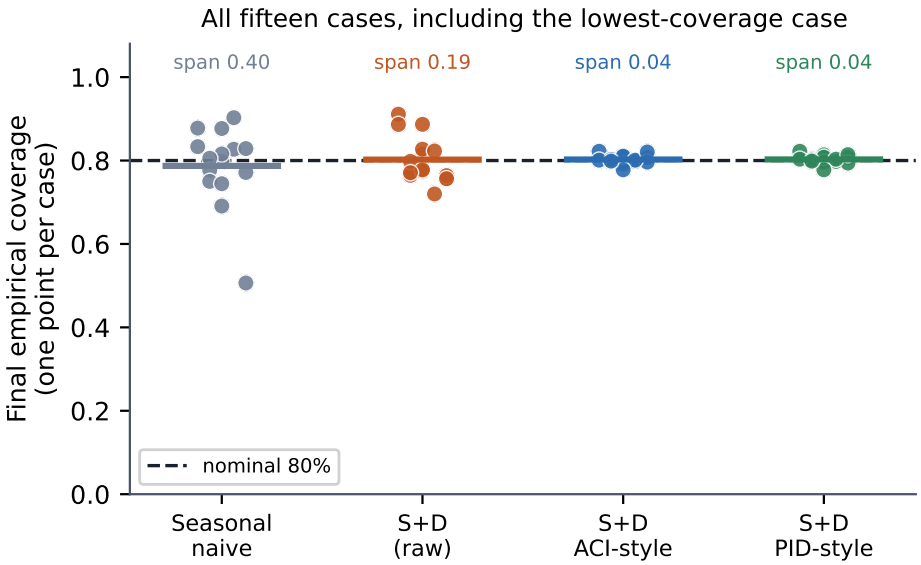


Figure 3: Final empirical coverage at nominal 80%, one point per case. The spread across cases is not a confidence interval or evidence of conditional coverage at each time.

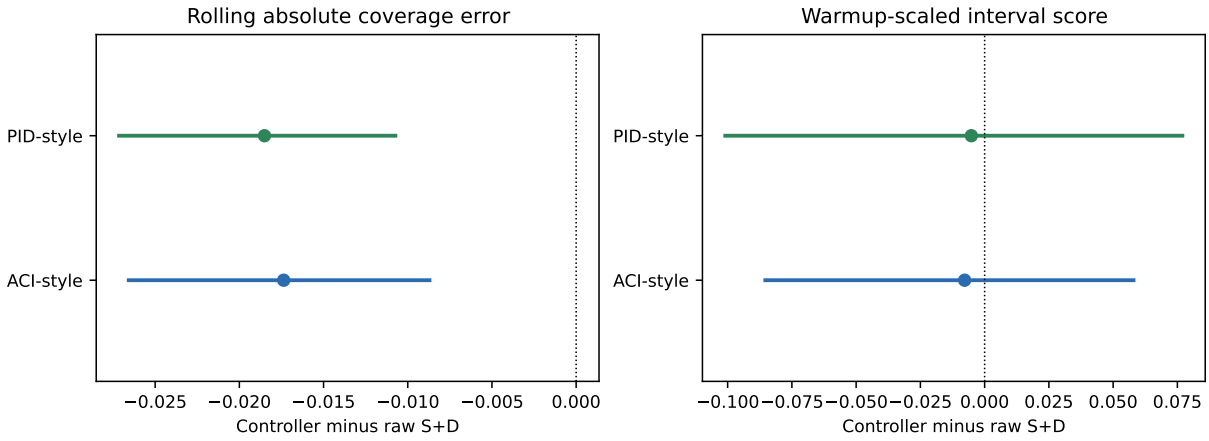


Figure 4: Controller-minus-reference differences with exploratory ninety-five-percent intervals from joint resampling of fifteen complete selected cases; the two panels use distinct dimensionless losses and horizontal scales.

Its PID-style counterpart is -0.01851, with interval [-0.02714,-0.01074]. The controllers strictly improve this metric on thirteen and fourteen of the fifteen streams, respectively. This agreement across most selected cases is stronger descriptive evidence than a difference in only the overall average. It remains evidence about this inventory: generated mechanisms, related sources and deliberately chosen excerpts do not acquire a population interpretation because resampling endpoints exclude zero.

The interval-score result is much less decisive. The ACI-style mean difference is -0.00778 with interval [-0.08533,0.05794], and the PID-style difference is -0.00517 with interval [-0.10092,0.07694]. Only five and six cases have strictly lower interval score. Thus the small aggregate score decrease is not a uniform case-level improvement, and both exploratory intervals include zero. The evidence supports a favorable rolling-coverage description while leaving a much weaker claim about total interval quality. It would be misleading to describe the coverage reduction as an equally strong reduction in every proper score.

Width and coverage can move together in ways a single summary conceals. In these results the average scaled width is slightly lower for both controllers than for raw S+D, but that average does not show how each interval reacts around a break or an unusually large residual. The per-update archive permits that inspection. The present paper reports its finite summary and sensitivity rather than selecting attractive local episodes as independent trials. It also retains the seasonal-naive reference, whose different point path means its score difference cannot be attributed solely to interval adaptation.

The paired analysis independently recalculates hits from the saved endpoints, rolling means through prefix sums, and interval penalties from the saved targets. It refuses disagreement with the retained hit column or an unavailable warmup scale and verifies identical S+D point paths. Independent hand examples check overlapping windows and a violated interval; positive affine source transformations preserve the scaled result. These are numerical validation checks. They make the new figure's computation auditable without proving representative sampling, arbitrary-sequence calibration or operational value.

7 Ablation and sensitivity

Table 6 retains the full fixed gain range, and Table 7 keeps generated realizations separate from real excerpts. Sensitivity retains all 630 case-setting rows. At nominal 90% on the S+D base,

Table 6: Mean rolling absolute error across fifteen cases. Controller cells give the range of the three fixed gain multipliers (0.5, 1, 2), not sampling uncertainty or a selected optimum.

Base	c	Raw	ACI-style	PID-style
Seasonal naive	0.7	0.1543	0.0976–0.1211	0.0907–0.1176
Seasonal naive	0.8	0.1309	0.0849–0.1033	0.0774–0.1016
Seasonal naive	0.9	0.0957	0.0629–0.0766	0.0596–0.0731
S+D	0.7	0.1085	0.0712–0.0864	0.0667–0.0852
S+D	0.8	0.0830	0.0597–0.0697	0.0558–0.0667
S+D	0.9	0.0567	0.0499–0.0571	0.0494–0.0557

one ACI-style gain setting has slightly higher rolling absolute error than the raw interval, so improvement is not uniform across settings. A further calculation regenerates each of the eleven synthetic mechanisms at seeds 43 and 44, keeping its definition and controller settings fixed. Together with the original seed 42, these give three realizations per generated mechanism, not 33 independent real-world datasets. The four real excerpts have no artificial repeated seed and are not counted in the generated-only table. These analyses assess numerical sensitivity of the specified design; they are not prospective validation on an independently selected collection.

Table 7: S+D base at nominal 80%, default gains: mean rolling absolute error on eleven generated mechanisms for each seed. Seeds share generator definitions; real series are excluded from this table.

Seed	Raw	ACI-style	PID-style
42	0.0684	0.0541	0.0525
43	0.0689	0.0580	0.0555
44	0.0693	0.0609	0.0587

Figure 5 makes each fixed gain cell visible instead of reporting only the minimum of its range. The raw reference does not depend on a controller gain, so it appears as a horizontal line in each panel. The two rows retain different point bases; the three columns retain different nominal coverages. This layout permits direct inspection of the S+D ninety-percent setting where one ACI-style cell has slightly higher rolling error than the raw interval. An average improvement at eighty percent therefore does not establish improvement at every neighboring parameter value.

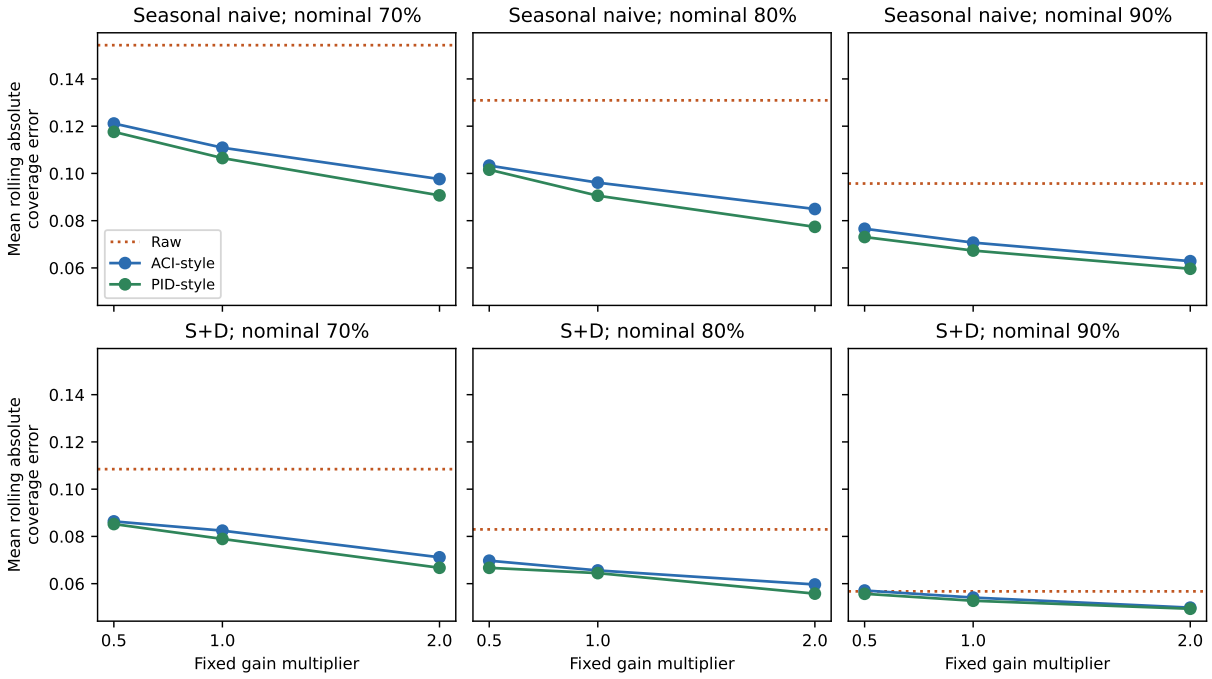


Figure 5: All fixed gain multipliers for both point bases and three nominal coverages, with equal weighting of the same fifteen streams; lines describe sensitivity rather than fitted response surfaces.

Figure 6 shows final coverage for every generated mechanism at the three retained seeds. It complements Table 7, which averages rolling error across mechanisms. Neither a favorable mean nor a small range of mechanism averages establishes identical behavior within each mechanism. The seeds change random realizations while retaining the same generator family and settings, so this is a structured sensitivity exercise. The four real excerpts appear in the original case comparison and receive no artificial seed labels.

The available ablations separate the point-base choice, presence of a controller and a common multiplier on its gains. They do not isolate every internal component. In particular, the PID-style grid multiplies proportional, integral and difference coefficients together. It is not a factorial ablation of those three terms, and no claim that one term causes the gain follows from these tables. Buffer capacity and quantile-interpolation convention also remain fixed. A future component study should specify those alternatives and its selection rule before using a new evaluation

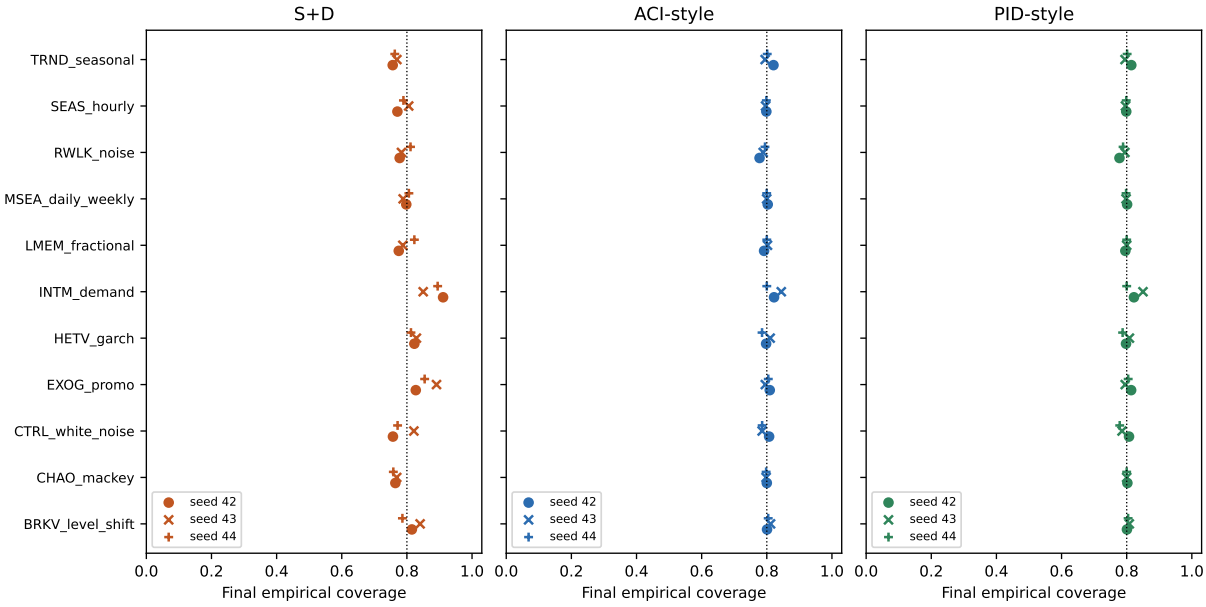


Figure 6: Final empirical coverage for eleven generated mechanisms at seeds forty-two, forty-three and forty-four; markers retain mechanism identity and are not independent samples from an authentic-data population.

collection. The current figures retain the historical design and its observed limitations without presenting unrun alternatives as measured results.

8 Discussion

The principal empirical result is lower rolling absolute coverage error for the two specified controllers on most of the selected streams. It is narrower than either a new conformal theorem or a general claim about forecasting architecture. The interval rules respond to accumulated past misses, while their shared point model remains unchanged. This mechanism makes a coverage improvement plausible, but the proper-score comparison shows why plausibility should not be mistaken for complete uncertainty improvement. Slightly lower aggregate interval score coexists with worse scores on a majority of cases.

The difference between the two coverage summaries also changes the interpretation of magnitude. The previously emphasized sevenfold ratio uses a statistic that averages signed rolling deviations before applying the absolute value. Its improvement can reflect balancing undercoverage and overcoverage at different times. Averaging absolute rolling deviations instead yields reductions near one fifth on these streams. Neither number is erroneous when correctly defined; the error would be to use the smaller statistic's ratio as if it described uniformly accurate local coverage. The deterministic inequality and rolling-boundary weights explain that distinction without relying on a significance test.

The sensitivity grid identifies dependence on the base and nominal level. It is useful because a controller tuned around one error distribution can behave differently around another. It cannot establish a generally optimal gain, especially when the original gain rationale is unavailable and the same streams are used for every cell. A stronger follow-up would prospectively fix authentic-data collections, identify a separate tuning budget, and include faithful published adaptive and score-predictive comparators. AgACI, strongly adaptive methods, decaying-step procedures and SPCI address relevant design questions, but none is numerically evaluated by the historical controller labels.

For research and teaching, the practical value of the replay is its inspectable information order and complete numerical state. A reader can locate the forecast made before a target arrived, reconstruct its buffer, separate its point error from interval changes, and check the chosen coverage statistic. For an operational forecasting service, additional questions remain: the cost of native updates, delayed or revised targets, missing observations, covariate arrival, simultaneous horizons and a loss attached to decisions. The current one-step full-prefix replay neither answers nor rules out those extensions. Its bounded findings provide a reproducible reference from which those better-specified experiments can proceed.

9 Limitations and interpretation

The study has one-step forecasts, short selected streams, two lightweight base constructions, three nominal levels and three gain multipliers. The bases share Gaussian dispersion approximations. There is no native stateful forecasting comparison, faithful published Conformal PID comparator, population sampling design, dependence-aware significance claim or operational cost study. A parameter range explored on these same cases cannot establish robustness outside it.

The numerical reproduction closes the missing per-update forecast and controller-state gap for the reconstructed experiment. It cannot retroactively supply a rationale for the original gains, the original machine's environment, or unavailable upstream acquisition revisions. Input hashes identify the exact redistributed values; they do not establish that those values represent all source-dataset populations. Historical measured latencies are therefore excluded from numerical-performance conclusions.

Improved empirical coverage in this construction is compatible with substantial local error, wider intervals or other unmeasured failure modes. The selected negative controls and information-order tests support bounded checks; they do not prove freedom from every possible leak. The appropriate conclusion is lower measured rolling coverage error for these particular controllers and streams, with interval efficiency and sensitivity reported alongside it.

Internal validity is limited by reconstruction rather than original prospective instrumentation. The procedure rederives every prediction and update using source-bound values and verifies archived outputs, but original per-call process state and acquisition metadata were not fully retained. The replay's exact numerical matches do not establish a recovered historical software environment in every respect. Realistic delayed observations, revised measurements and missing targets would require an explicit arrival policy beyond the immediate one-target update used here.

External validity is limited by the deliberate selection of mechanisms and excerpts. The additional generated seeds exercise random variability within those definitions. They do not test new source populations, multivariate models, long-horizon joint events or the pretrained forecasters available elsewhere in the application. The study also lacks a faithful comparison with the published adaptive methods reviewed above. Its title and conclusions consequently remain tied to fifteen-series replay, while independent reader review of this working revision remains pending. Publication acceptance is not inferred from successful compilation or numerical reconstruction. These engineering checks and scientific limitations must remain separately visible in any later release description.

10 Reproducibility and availability

The source, full numerical replay, sensitivity rows, input hashes and deterministic figure/table generators are at `https://github.com/fsantibanezleal/CAOS_RES_ChronoScope`. In a de-

clared product environment, use Python to run `scripts/manuscript_replay.py` with `-check`, then run `scripts/make_manuscript_tables.py`. The streaming manuscript directory contains the numerical replay as `evidence/replay.json`. Source values are included for each case; raw upstream acquisition histories and original timing metadata are not reconstructed. The separately published `preqts==0.3.0` package supplies the checked historical controller implementation. Code and redistributed data retain their stated licences; prose uses the licence in the header.

The additional analysis is generated with `scripts/analyze_replay_cases.py`; its `-check` option verifies the committed numerical extension and `-render` rebuilds its three figures and paired table. Its source digest is fixed in `next/evidence/analysis-plan.json`. This analysis does not overwrite the published replay or its original tables. The working manuscript retains a separate revision status, coverage map and unpublished designation until its numerical, content and independent-review requirements are satisfied.

11 Conclusions

On fifteen selected streams at nominal eighty-percent coverage, the historical ACI-style and PID-style buffer controllers lower mean rolling absolute coverage error from .0830 to .0656 and .0645. The reduction is smaller than the ratio obtained by averaging signed deviations before taking their absolute value. Reconstructed per-update records, explicit score definitions and a deterministic inequality show why those statistics answer different questions.

Retrospective joint case resampling gives rolling-error differences of -.01737 and -.01851, with exploratory intervals excluding zero for this case-weighting exercise. The corresponding scaled interval-score differences are close to zero, their intervals include zero, and most cases do not improve that score. This limits the empirical conclusion to the specified coverage behavior rather than a general advantage in probabilistic prediction.

The replay preserves its exact bases, numerical floors, clipping and inherited gains. It does not evaluate native state updates or faithfully reproduce published Conformal PID Control. Completing those comparisons, independent reader review and prospectively designed authentic-data studies are future requirements. The current working revision documents a reproducible finite evaluation and its limits; it is not a new coverage guarantee or an accepted deployment policy.

References

[1] V. Vovk, A. Gammerman and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005. `https://doi.org/10.1007/b106715`.

[2] G. Shafer and V. Vovk, A tutorial on conformal prediction. *JMLR*, 9, 371–421, 2008. `https://jmlr.org/papers/v9/shafer08a.html`.

[3] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani and L. Wasserman, Distribution-free predictive inference for regression. *JASA*, 113, 1094–1111, 2018. `https://arxiv.org/abs/1604.04173`.

[4] Y. Romano, E. Patterson and E. Candès, Conformalized quantile regression. *NeurIPS*, 2019. `https://proceedings.neurips.cc/paper/8613-conformalized-quantile-regression.pdf`.

[5] R. F. Barber, E. Candès, A. Ramdas and R. J. Tibshirani, The limits of distribution-free conditional predictive inference. *Information and Inference*, 2021. https://arxiv.org/abs/1903.04684.

[6] V. Chernozhukov, K. Wüthrich and Y. Zhu, Exact and robust conformal inference methods for predictive machine learning with dependent data. COLT, 2018. https://proceedings.mlr.press/v75/chernozhukov18a.html.

[7] R. J. Tibshirani, R. F. Barber, E. Candès and A. Ramdas, Conformal prediction under covariate shift. *NeurIPS*, 2019. https://proceedings.neurips.cc/paper_files/paper/2019/hash/8fb21ee7a2207526da55a679f0332de2-Abstract.html.

[8] R. F. Barber, E. Candès, A. Ramdas and R. J. Tibshirani, Conformal prediction beyond exchangeability. *Annals of Statistics*, 51, 816–845, 2023. https://arxiv.org/abs/2202.13415.

[9] C. Xu and Y. Xie, Conformal prediction interval for dynamic time-series. ICML, 2021. https://proceedings.mlr.press/v139/xu21h.html.

[10] C. Xu and Y. Xie, Sequential predictive conformal inference for time series. ICML, 2023. https://proceedings.mlr.press/v202/xu23r.html.

[11] M. Zaffran, O. Féron, Y. Goude, J. Josse and A. Dieuleveut, Adaptive conformal predictions for time series. ICML, 2022. https://proceedings.mlr.press/v162/zaffran22a.html.

[12] I. Gibbs and E. Candès, Conformal inference for online prediction with arbitrary distribution shifts. *JMLR*, 25, 1–36, 2024. https://jmlr.org/papers/v25/22-1218.html.

[13] A. Bhatnagar, H. Wang, C. Xiong and Y. Bai, Improved online conformal prediction via strongly adaptive online learning. ICML, 2023. https://proceedings.mlr.press/v202/bhatnagar23a.html.

[14] A. N. Angelopoulos, R. F. Barber and S. Bates, Online conformal prediction with decaying step sizes. ICML, 2024. https://proceedings.mlr.press/v235/angelopoulos24a.html.

[15] T. Gneiting, F. Balabdaoui and A. E. Raftery, Probabilistic forecasts, calibration and sharpness. *JRSS B*, 69, 243–268, 2007. https://doi.org/10.1111/j.1467-9868.2007.00587.x.

[16] L. J. Tashman, Out-of-sample tests of forecasting accuracy: an analysis and review. *International Journal of Forecasting*, 16, 437–450, 2000. https://doi.org/10.1016/S0169-2070(00)00065-0.

[17] R. J. Hyndman and A. B. Koehler, Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22, 679–688, 2006. https://robjhyndman.com/papers/mase.pdf.

[18] B. Efron, Bootstrap methods: another look at the jackknife. *Annals of Statistics*, 7, 1–26, 1979. https://doi.org/10.1214/aos/1176344552.

[19] H. R. Künsch, The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17, 1217–1241, 1989. https://doi.org/10.1214/aos/1176347265.

[20] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, third edition. OTexts, 2021. https://otexts.com/fpp3/exponential-smoothing.html.

[21] A. Trindade, ElectricityLoadDiagrams20112014. UCI Machine Learning Repository, 2015. https://doi.org/10.24432/C58C86.

[22] S. Chen, Beijing PM2.5. UCI Machine Learning Repository, 2015. https://doi.org/10.24432/C5JS49.

[23] R. Godahewa, C. Bergmeir, G. I. Webb, R. J. Hyndman and P. Montero-Manso, Monash Time Series Forecasting Archive. NeurIPS Datasets and Benchmarks, 2021. https://arxiv.org/abs/2105.06643.

[24] A. P. Dawid, "Statistical theory: The prequential approach," *JRSS A*, vol. 147, no. 2, pp. 278–292, 1984, doi:10.2307/2981683.

[25] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," *NeurIPS*, 2021, https://arxiv.org/abs/2106.00170.

[26] A. N. Angelopoulos, E. Candès and R. J. Tibshirani, "Conformal PID control for time series prediction," *NeurIPS*, 2023, https://arxiv.org/abs/2307.16895.

[27] R. J. Hyndman and B. Billah, "Unmasking the Theta method," *International Journal of Forecasting*, vol. 19, no. 2, pp. 287–290, 2003, https://robjhyndman.com/papers/Theta.pdf.

[28] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *JASA*, vol. 102, no. 477, pp. 359–378, 2007, doi:10.1198/016214506000001437.